Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

SUMMARY

Senior Software Engineer, ***8 years***: ***AI inference serving***, ***backend microservices***, cloud-native infrastructure at ***Amazon, Microsoft, Oracle, Razorpay***. Strong ***Python***, ***modern memory-safe C++***, ***Rust***, ***CUDA***; ***BERT inference at 5,000 QPS*** (ONNX, INT8) on ***Docker/Kubernetes*** with CI/CD; passionate about ***CV/DL for road safety*** on ***NVIDIA Jetson Orin*** (TensorRT).

PERSONAL PROJECTS

Driving-CivicSense: Full-Stack Edge AI (Rust, CUDA), NVIDIA Jetson Orin Nano Super NX | github

- ***Full-stack data collection***: ***50,000 miles*** of driving footage across all weather and terrain via vehicle camera sensors and ***LiDAR***; simulation-driven synthetic data; privacy-first, on-device processing on edge neural chips.
- ***Multi-chip edge system***: ***YOLOv8/v11 INT8 → TensorRT*** at ***~ 12 ms*** on ***Jetson Orin Nano Super NX***; sensor data streamed to a ***Raspberry Pi Zero 2 W UDP server***; validated on ***Raspberry Pi 5*** and ***Orange Pi Zero***.
- ***Containerized VLM inference***: building edge VLM serving with ***vLLM*** and ***TensorRT-LLM*** in Docker containers on the Jetson.
- ***CUDA-Oxide (Rust-to-CUDA)***: contributing GPU kernels and documentation fixes (unsafe intrinsics wrapped in explicit unsafe blocks); pushing Jetson limits with idiomatic, memory-safe, fearlessly concurrent kernels.
- ***Research & safety standards***: paper on ***Intersection Blockage Prediction*** | paper; studying ISO 26262. Books: ***Seeing Machines*** · ***CUDA Kernels*** | book

EXPERIENCE

Software Developer II, Microsoft, Redmond, WA — Jun 2025 – Aug 2026

- ***Containerized Cloud Infrastructure***: zero-trust sovereign-cloud networking with eBPF + Cilium enforcing ***data residency*** & SecNumCloud compliance; Kubebuilder operators reconcile state across AKS clusters (France, Germany).
- ***Systems Performance***: owned the ***C++/C# → Rust*** migration of Microsoft Information Protection; eliminated use-after-free/ buffer-overflow/null-safety bugs; ***~ 40% lower P99***, ***2x throughput***, ***3x lower memory***, six-figure savings.
- ***Automation***: AI engine auto-generating Cilium policies and scanning misconfigurations; review ***4 hours → near-real-time***, zero drift.
- *TechStack*: ***Rust, C++, Python, eBPF, XDP, Cilium, Kubernetes (AKS), Docker, CI/CD, Linux***

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA — Oct 2024 – Mar 2025

- ***Backend Services & SDKs***: owned the ***Terraform provider*** (developer SDK) for Autonomous Database, replacing legacy control-plane tooling; ***$1.2M/month*** projected savings.
- ***High-Performance Systems***: lock-free data structures cutting contention in high-concurrency read-write paths; ***+ 45,000 ops/s*** .
- ***Security & Key Management***: integrated OCI Security Vault with automated key rotation across control-plane services.
- *TechStack*: ***Java, Go, REST/gRPC, Terraform, OCI***

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India — Mar 2021 – Oct 2024

- ***Production Inference Serving***: scaled low-latency ***BERT-based inference*** to ***5,000 QPS*** (ONNX Runtime, INT8 quantization, latency tuning); validated throughput/latency; cut triage SLA ***7 days → 2 hours*** .
- ***Real-Time ML Ranking***: owned end-to-end (training, serving, latency tuning) XGBoost + Deep Learning ensemble across ***FreeVee, Twitch, MiniTV, Prime Video, Amazon Retail***; ***5% ad CTR lift***, ***$5.4M/month revenue***.
- ***Event-Driven Microservices***: led a team of 3 building ***programmatic guaranteed (PG) ad delivery*** at ***< 50 ms*** via real-time CDC (Kotlin); REST/gRPC APIs.
- ***Distributed Systems***: designed a uniqueness-constraint indexing SDK (***two-phase commit***, ***serializable isolation***, ***3K writes/s***).
- ***Data Platform***: built the ***1-petabyte*** warehouse/lake (AWS Glue, Spark) for financial reporting & pay computation.
- ***Telemetry & Automation***: real-time monitoring and automated controls cut deal over-delivery ~ ***30%*** and budget overspend ***25%*** .
- *TechStack*: ***Python, Kotlin, PyTorch, XGBoost, ONNX Runtime, Spark, Kafka, DynamoDB, Redshift, AWS***

SDE, Razorpay, Bengaluru, India — Aug 2020 – Feb 2021

- ***Backend Microservices***: P2P Unified Payments Gateway (Go, Protobuf/gRPC): ***3M+ transactions/hour*** peak, ***99.99% uptime***, ***150ms P99***; customers: ***Zomato, Swiggy, Zerodha, Groww*** (AWS, GCP).

Software Engineer, Mindfire, Bhubaneswar, India — Aug 2018 – Feb 2020

- Java/Spring full-stack tools; AR product-search microservice (***10K+ daily queries***).

SKILLS

Languages: Python, C++, Rust, Go, Java, Kotlin, C, Shell/Bash
AI/ML Inference: Deep Learning, Computer Vision, PyTorch, XGBoost, BERT, LLMs, ONNX Runtime, TensorRT (INT8), vLLM, SGLang, TensorRT-LLM, YOLOv8/v11, Edge Inference, Quantization, MLOps
GPU & Parallel Computing: CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming, NVIDIA Jetson Orin, HPC
Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Performance Engineering, Low-Latency Systems, Computer Architecture
Networking: TCP/IP, mTLS, HTTP/2/3, QUIC, eBPF, XDP, Cilium, Load Balancing
Distributed Systems & Streaming: Consensus, Replication, Sharding, Kafka, Spark, CDC, Microservices, gRPC, REST, Protobuf
Cloud & Infra: AWS (EC2, ECS, EKS, Lambda, Kinesis, EMR, SageMaker, CDK), Azure Kubernetes Service (AKS), Google Kubernetes Engine (GKE), Kubernetes, Docker, Helm, Terraform, Kubebuilder, Istio, CI/CD
Observability: Prometheus, Grafana, AWS CloudWatch
Data & Storage: DynamoDB, PostgreSQL, MySQL, Redis, Amazon Redshift, Amazon S3

EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018